

SHREYA PITHVA

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About

I am interested in theoretical physics, especially the connection between quantum field theory and quantum computing. I enjoy learning the core concepts of quantum physics and gradually building a deeper understanding about the fundamentals of Universe. I want to explore open questions in the field as I continue to grow academically.

Education

Current Position: Master of Science (2026-27) at Indian Institute of Science (IISc)

Bachelor of Science | Major: Physics, Minor: Quantum Technology Completed (8th Semester)
Indian Institute of Science (IISc) Bengaluru (2022-2026)
Current GPA: 8.33/10

Class 12 (CBSE Central board of Secondary Education, India) Year: 2022
Silver Bells Public School Bhavnagar, Gujarat
Overall percentage: 94.4 %

Class 10 (CBSE Central board of Secondary Education, India) Year: 2020
Silver Bells Public School Bhavnagar, Gujarat
Overall percentage: 90.2 %

Honours And Awards

QC RISE Internship Program (Sep 2026 – Feb 2027) Year: 2026

- Selected for QC RISE (under National Quantum Mission) fellowship in rapidly advancing field of Quantum Computing.

S.N Bhatt Memorial Excellence Fellowship 2026 Year: 2026

- Selected for the ICTS–S. N. Bhatt Memorial Excellence Fellowship (2026), a competitive research program at the International Centre for Theoretical Sciences (TIFR).

Mallika science fellowship for four years (2022-26) of B.S. (Research). Year: 2022

- For female students with good rank in highly competitive exams of the country.

International Astronomy and Astrophysics Competition (IAAC),

- Achieved **top 7%** ranking internationally in the final round of the **IAAC** awarded **Silver Honour**. Year: 2024

International Physicists Tournament (IPT) Year: 2024

- Obtained Rank 3 in India at the **International Physicists Tournament (IPT)** national selections.
- Wrote Paper along with 5 teammates on “Varying parameters (e.g., peg size, distribution, bead format in Galton Board) and introducing vibrations can yield distributions that deviate from binomial and may not adhere to the central limit theorem in an independent and identical distributed scenario”.
- Learnt 3D modelling and 3D printing.
- Represented IISc at national level.

IIT JEE ADVANCED - 2022 All India Rank 3772 Year: 2022

Indian Institute of Technology – Joint Entrance Examination (IIT-JEE)
Joint Admission Board (JAB), Government of India

- Applicants: Approximately 155,538 students (JEE MAIN Qualified students out of 1,026,799 students)
- IIT-JEE of India is listed as the second in the toughest exam in the world’s list.

JEE MAIN Percentile Rank - 98.87 % Year: 2022

National Testing Agency (NTA), Government of India

- Secured 98.875 percentile Rank.
- Applicants: Approximately 1,026,799 students.

LIC Student of the Year Award in 11th standard Year: 2021

Workshops and Conferences

- Participated and Volunteered for TQC 2025 Conference (Theory of Quantum Computation, Communication and Cryptography (TQC)) held at IISc, Bengaluru.

- **Selected for poster presentation in ASI 2025 (Astronomical Society of India) conference.**
Abstract “Systematics of the Multi-Frequency Properties of Accreting Supermassive Black Holes in the Local Universe”
<https://ui.adsabs.harvard.edu/abs/2025asi..confP..97P/abstract>, https://astron-soc.in/asi2025/abstractdetails/ASI2025_89
- Selected and attended “**Summer School in Optics and Photonics'24**” at IISc.
- Selected for Summer School for Women in Physics, 2024 at **ICTS (International Centre for Theoretical Sciences, Bengaluru)** from 3rd June to 14th June 2024.
(one of the 65 students to be selected in the entire country (India))

“**Beginning Astronomy v3: Start a data-driven journey**” workshop held on 11-13 Dec 2023 at IIT Hyderabad by IUCAA.

- one of the 35 students to be selected in the entire country (India)
- learnt to access telescopes, satellite, and Solar Dynamics Observatory data and to analyse it using Python and SAO Image DS9 software.
- Learnt about solar physics and formation of sunspots, Pulsar, Neutron stars, LIGO detection of gravitational wave.

Certificates /Online Courses

- “**From Big Bang to Dark Energy**” by the University of Tokyo (organised by Hitoshi Murayama).
- Certificate course by **MATLAB** Onramp
- “**Elements of AI**” by University of Helsinki (2 ECTS credits).
- “**How Things work: An Introduction to physics**” by the University of Virginia
- Qualified the **SCRATCH assessment by Google**, conducted as a part of Google India Code (2018 training program).
- **Certificate by ISRO** for participating in an exhibition and explaining about rockets and satellite to the public.

Student Membership

- **QUARKS** Magazine (Ug club) – [Core Member](#) Year: 2026
- Convenor of **ASTRAE CLUB (Indian Institute of Science** Astronomy club). Bengaluru, Year:2024
(Organise Talks, Night Sky observation Sessions using Telescopes, Astrophotography)
- Core member of UG Magazine club (QUARKS) Year: 2026
- APS American Physical Society Year:2024
- Space Generation Advisory Council. Year:2023
- SSPI UK (Space and Satellite Professionals International - UK Chapter). Year:2023
- Space Crew Working Group (Innova Space). Year:2023
- LIFE-TO and beyond. Year:2023

Previous Projects

- Reading project on MOND modified Newtonian dynamics under **Professor Nirmal Raj at IISc**.
(Modified Newtonian Dynamics (MOND) is an alternative to dark matter that explains galaxy rotation curves by modifying Newton's second law at low accelerations. It introduces a characteristic acceleration scale, a_0 below which gravitational forces deviate from the Newtonian form. While MOND accounts for flat galaxy rotation curves, it faces challenges in explaining large-scale cosmological observations.)
- Nonlinear photonics and Raman Fibers (Reading project and some practical’s experiments) under Professor **Superdeep at IISc**.
- On Super Massive Black holes under Professor **Prajval Shastri at RRI (Raman Research Institute)**.
(I was engaged in a computational analysis with extensive literature review, investigating the dynamics of accreting supermassive black holes in distant galaxies. My focus spans multiwavelength data analysis of S7 survey, exploring how these black holes impact star formation within their host galaxies, particularly studying synchrotron – emitting jets and their interplay with the nuclear environment.)
- As part of the QFT on QC course taught by **Prof. Aninda Sinha**, I worked with a M.Tech student to study light–matter interaction using the Jaynes–Cummings model and Rabi model. We simulated the system on quantum computers using Qiskit and compared the results with analytical calculations in Wolfram Mathematica. The project investigated phenomena such as energy level splitting and photon absorption and emission, and explored regimes where standard approximations break down. *(Grade Points: 74.67/100)*
- **BS thesis** work under **Prof. Aninda Sinha**, I studied scalar ϕ^4 theory in (1+1) dimensions using tensor-network methods, including DMRG and iDMRG with TeNPy. I studied finite and infinite-chain formulations, convergence with bond dimension, and local bosonic truncation.

On Going Projects

- **S. N. Bhatt Fellowship Project, under Prof. Abhishek Dhar at ICTS**
Study of quantum open systems, focusing on nonequilibrium dynamics, decoherence, and system–environment interactions.
- **Master’s Thesis:** Quantum Scrambling algorithm under **Prof C.M. Chandrashekar**
Research interest includes quantum scrambling, the Hayden-Preskill protocol, and black-hole information.

Preprint

Dhruv Shah, R.K. Shishir, Manjaree, Shreya Pithva, TY Boorith Balaji, Rahul Agarwal Singh. Non-Gaussian statistics in static and dynamics Galton Boards. (also linked to NASA ADS)

<https://doi.org/10.48550/arXiv.2404.19478>

Skills

- Efficient in Python, SQL, MATLAB, Mathematica, LATEX
- Comfortable with QISKIT and QuTip
- Efficient in telescope handling (Dobsonium, Cassegrain, Maksutov)
- Familiar with the basics of designing 3D models (in Fusion 360 software) and 3D printing.
- Learned astrophotography and image processing using Siril Software.

Teaching & Academic Outreach

Physics and Mathematics Tutor, QUAZAR
(<https://www.quazaredu.com/>)

February 2026 – Present

- Tutor students in physics and mathematics, with a focus on building conceptual understanding and problem-solving skills.
- Mentor students preparing for competitive examinations and olympiads, including NSEP and Physics Olympiad-level preparation.
- Occasionally volunteer on Saturdays, when time permits, to teach English to children in government schools as part of IISc’s Notebook Drive outreach initiative.

Co-curricular Activities & Interests

- **Reading:** I Enjoy literary fiction, contemporary novels, and psychological thrillers, including works such as *A Thousand Splendid Suns*, *A Little Life*, and *The Silent Patient*. ([goodreads](#))
- **Trekking and Travel:** Passionate about travelling and trekking; completed several treks, including moderate-to-difficult Himalayan routes such as the Kartik Swami Temple trek and the Dhano 2/Kugti Pass route in Himachal Pradesh (India).
- **Snooker:** Senior snooker player; won multiple medals in IISc tournaments, including undergraduate and overall-category events.
- **Writing:** Occasionally write [blogs](#) and [poems](#); core member of [QUARKS](#), the IISc undergraduate magazine club.
- **Painting and Sketching:** Practising [canvas painting](#) and sketching since childhood; won state-level and other art competitions.

Languages

- English - FLUENT
- Hindi - FLUENT
- Gujrati – NATIVE